

Mathematical Methods

B. Grey Vandenberg

1 Real Numbers

A *geometric sequence* $a_1, a_2, \dots$ is one where each successive term in the sequence is obtained via multiplication by a common factor on the most recent term. For example, each of the following is a geometric sequence:

$$2, 4, 8, 16, 32, \dots \quad (1.A1)$$

$$1, \frac{2}{3}, \frac{4}{9}, \frac{8}{27}, \frac{16}{81}, \dots \quad (1.A2)$$

$$a, ar, ar^2, ar^3, \dots \quad (1.A3)$$

The general form of a geometric sequence given in the lattermost representation says that, if we start with the first term $a_1 \equiv a$, then each successive term $a_2, a_3, \dots$ is obtained via multiplication by the common factor r , which we call the *common ratio* of the sequence $\{a_n\}_{n=1}^\infty$. (We mention the notation: in general, we use a_k with the index $k \in \mathbb{N}$ taking values in the positive integers to represent the k th term in the sequence. Thus, a sequence $a_1, a_2, \dots$ may be expressed by, for example: $\{a_n\}_{n=1}^\infty$, $\{a_k\}_{k \in \mathbb{N}}$, $\{a_1, a_2, \dots\}$, and so on. As we continue, we may even simply write a_n with the understanding that the index n typically means “an index that varies over the natural numbers $\mathbb{N}$.” None of these notations should be entirely surprising, and we use each freely.)

There is a clear distinction between a sequence like (1.A1), where the terms get larger and larger as we progress, compared to one like (1.A2) whose terms get smaller and smaller. In the notation of (1.A3), we observe that sequence (1.A1) has $a = 2$ with $r = 2$, while (1.A2) has $a = 1$ and $r = 2/3$ (check this!). It is immediately clear that the growth in (1.A1) is a result of the multiplication by $r = 2 > 1$, while the shrinkage in (1.A2) results from the multiplication by $r = 2/3 < 1$.

Sequences on their are of major interest when studying algorithmic procedures — for instance, if we claim that some numeric optimization method produces smaller and smaller error as we continue to iterate, we may want to test the performance of our algorithm against known test-cases whose analytic results are known. In such a case, we may call the error at the n th iteration of the algorithm $\epsilon_n = |a - a_n|$, where a_n would represent the numeric result produced by our algorithm at the n th iteration, while a is the analytic result to which we hope our algorithm *converges* as we continue to iterate. If we are confident that *eventually* it will be the case that the algorithm converges to the value a , we might say that a is the *limit* of the sequence a_n , and write

$$a := \lim_{n \rightarrow \infty} a_n. \quad (1.A4)$$

Relative to our $\epsilon_n = |a - a_n|$, it would follow that whenever (1.A4) holds, we would have $\epsilon_n \rightarrow 0$ as $n \rightarrow \infty$. (Note the two notations: if we were to call $\epsilon := 0$, then we could write $\lim_{n \rightarrow \infty} \epsilon_n = \epsilon$.)

It can often feel a bit more natural to write “the sequence $\epsilon_n \rightarrow \epsilon$ as $n \rightarrow \infty$,” which omits the $\lim$ symbol explicitly but remains implicit in the description. We may also write $a_n \xrightarrow{n \rightarrow \infty} a$ to denote the convergence of the sequence a_n to the limit a as the index $n \rightarrow \infty$).

On their own, sequences are useful bookkeeping devices. However, beyond the growth rate as we move down the line of a sequence, a major interest often lies in understanding its *collective behavior*. For instance, we understand from (1.A1) that the individual terms in the sequence $a_n \rightarrow \infty$ as $n \rightarrow \infty$. It may not be entirely clear how we could formalize this notion; to do so, we can make use of the following definition, which also formalizes some of the ideas from above.

Definition 1.0.1 (Sequential Convergence)

Let $\{a_k\}_{k \in \mathbb{N}}$ be a sequence. We say that a_k *converges to the limit* a if for any arbitrarily small $\epsilon > 0$, we can find a thresholding index $N \in \mathbb{N}$ such that for any $n \geq N$ indexing a member a_n at least as deep in the tail of the sequence as a_N , we have

$$|a_n - a| < \epsilon.$$

That is, say that a_n converges to the limiting point a if for any $\epsilon > 0$,

$$\exists N \in \mathbb{N}: \quad \forall n \geq N, \quad |a_n - a| < \epsilon. \tag{1.A5}$$

If no such limiting point a exists (i.e., if for any proposed limit a and any threshold N , we can find an index $n \geq N$ such that the distance $|a_n - a| \geq \epsilon$ for *some* $\epsilon > 0$), we say that the sequence a_n *fails to converge* (or that a_n is *nonconvergent*). If $a_n \rightarrow \pm\infty$ as $n \rightarrow \infty$, we say that a_n *diverges* (to $\pm\infty$, as appropriate).

The failure of a sequence $\{a_k\}_{k \in \mathbb{N}}$ to converge is loosely the property that for any proposed limit a , we can always find an $\epsilon > 0$ such that a_n fails to be ϵ -close to a at some point in the tail of the sequence. There are many types of failure we can imagine. Notice the subtle difference in the characterizations: a sequence converges if and only if for *any* $\epsilon > 0$, there exists $N \in \mathbb{N}$ such that $|a_n - a| < \epsilon$ for all $n \geq N$. On the other hand, to establish the converse, we only need to identify *some* $\epsilon > 0$ such that for any fixed index $N \in \mathbb{N}$, $|a_n - a| \geq \epsilon$ for *at least one* $n \geq N$. The point is that while a sequence which fails to converge can be quite easy to establish, convergence can be a bit trickier; the former needs us establish only one counterexample, while the latter needs a guarantee that holds for any $\epsilon > 0$.

Nevertheless, establishing the failure of a sequence to converge can be made simpler. In our characterization above, we said that a_n is a nonconvergent sequence if no limit point a exists. But as it stands, this would clearly require us to check every single limiting point a against our criterion; for instance, it seems somewhat clear that the sequence (1.A1) is nonconvergent, since the terms $2, 4, 8, 16, 32, \dots$ continue to get bigger and bigger. Intuitively, there is no way we could pick a single real number $a \in \mathbb{R}$ and find a member of the sequence $a_N \in \{2, 4, 8, 16, 32, \dots\}$ such that $|a_n - a| < \epsilon$ for all $n \geq N$ if we were to take, say, $\epsilon = 1$. It would thus seem immediately useful to establish a criterion that would allow us to characterize whether a sequence converges or not independently

of arbitrarily proposed limit points a . One way to do this is to notice that, if we look at *pairs of successive entries* (a_1, a_2) , (a_2, a_3) , and so on, the *pairwise distances* grow larger and larger as time goes on — in particular, $|a_2 - a_1| = |4 - 2| = 2$, $|a_3 - a_2| = 4$, $|a_4 - a_3| = 8$, and so on.

While a useful place to start, we must take some care in establishing this heuristic, because as stated it is too loose for some sequences whose terms exhibit a deceptively slow rate of decay. That is, even though

$$a_n \xrightarrow{n \rightarrow \infty} a \implies |a_{n+1} - a_n| = |a_{n+1} - a + a - a_n| \leq |a_{n+1} - a| + |a_n - a| \xrightarrow{n \rightarrow \infty} 0,$$

the converse *fails*. For instance, if we take¹

$$a_n = \log n,$$

then

$$|a_{n+1} - a_n| = |\log(n+1) - \log(n)| = \left| \log\left(\frac{n+1}{n}\right) \right| = \left| \log\left(1 + \frac{1}{n}\right) \right| \xrightarrow{n \rightarrow \infty} \log(1) = 0.$$

This is clear simply because the sequence $\{1/n\}_{n \in \mathbb{N}}$ has terms $1, 1/2, 1/3, 1/4, \dots$ converging to 0 as $n \rightarrow \infty$. In particular, given $\epsilon > 0$, if we propose the limit $a = 0$ with $a_n = 1/n$, then

$$|a_n - a| = \left| \frac{1}{n} - 0 \right| = 1/n \stackrel{\text{set}}{<} \epsilon \implies n > 1/\epsilon,$$

so that in order to get $a_n = 1/n$ within ϵ of the limit $a = 0$, we need simply look at the terms in the tail a_n for $n > 1/\epsilon$; any such term will satisfy the convergence definition (1.A5) relative to the given $\epsilon > 0$. This shows that, indeed, $a_n = 1/n \rightarrow 0$ as $n \rightarrow \infty$. It is thus certainly the case that

$$b_n := 1 + \frac{1}{n} \xrightarrow{n \rightarrow \infty} 1,$$

since clearly the addition of a constant 1 to the convergent sequence $1/n$ doesn't affect the convergence of $1 + 1/n$, and

$$c_n = \log\left(1 + \frac{1}{n}\right) \xrightarrow{n \rightarrow \infty} \log(1) = 0,$$

since taking the logarithm of numbers slightly larger than and consistently decaying toward 1 will result in logarithms which are slightly larger than and continuously decaying toward 0. Nevertheless, what we *also* know is that the natural logarithm is an *unbounded function*, in the sense that

$$\lim_{n \rightarrow \infty} \log(n) = \infty.$$

What this means is that if we look at the successive values of the original sequence $a_n = \log n$, we'll observe that $a_n \rightarrow \infty$ *even though* the pairwise distances $|a_{n+1} - a_n| \rightarrow 0$ as $n \rightarrow \infty$. All of this is to convince you that it is not sufficient to merely consider whether *pairwise distances* get small

¹Throughout, the symbol $\log$ is used to denote the natural logarithm of base e — i.e., we use $\log$ instead of $\ln$. This is largely because the base of the logarithm plays a less substantial role in limiting scenarios, as is apparent from the change of base formula — see Exercise XX.

and remain small in order to establish whether or not a sequence a_n converges to *some* finite limit. (Again, the interest at this point is *only* in “whether or not the sequence converges”; the question of “what it converges to” is the topic of future discussion).

All of this may cast doubt on the pursuit of a limit-agnostic criterion constructed relative only to properties of the sequence a_n itself. The trick, however, is that we should not ask whether *adjacent* terms eventually become close. We’ve observed that convergence is stronger than that: if a convergent sequence $a_n \rightarrow a$, then we imagine we can choose a point in the tail of the sequence a_N such that the distances $\epsilon_n := |a_n - a|$ get small and stay small for any $n \geq N$. Indeed, what’s clear is that if $a_n \rightarrow a$, then there exists an N such that $|a_n - a| < \epsilon$ for any $\epsilon > 0$. But since this has to hold for *any* such $\epsilon > 0$, we can just as well force it to hold for $\epsilon' := \epsilon/2$ by finding an appropriate index N' . In this case, the triangle inequality implies that for any two indices $n, m \geq N'$, each of $|a_n - a| < \epsilon' \equiv \epsilon/2$ and $|a_m - a| < \epsilon' \equiv \epsilon/2$, so that

$$\forall n, m \geq N': |a_n - a_m| = |a_n - a - (a_m - a)| \leq |a_n - a| + |a_m - a| < 2\epsilon' \equiv \epsilon.$$

To summarize, what we’ve just established is that if $a_n \rightarrow a$, then for any $\epsilon > 0$, we can identify a thresholding index $N \in \mathbb{N}$ such that the distance between *any two* points further in the tail a_n and a_m , $n, m \geq N$ gets small and stays small:

$$a_n \xrightarrow{n \rightarrow \infty} a \implies \text{Given any } \epsilon > 0 : \left(\exists N \in \mathbb{N} : \forall n, m \geq N, |a_n - a_m| < \epsilon \right). \quad (1.A6)$$

We now state the formal characterization of this phenomenon.

Definition 1.0.2 (Cauchy Sequences)

A sequence $\{a_n\}_{n \in \mathbb{N}}$ is said to be a *Cauchy sequence* if for any $\epsilon > 0$, there exists a threshold $N \in \mathbb{N}$ such that for all $n, m \geq N$,

$$|a_n - a_m| < \epsilon.$$

Intuitively, we can imagine that a Cauchy sequence $\{a_n\}_{n \in \mathbb{N}}$ is one where no matter how small the window $\epsilon > 0$, we can always find a point in the tail of the sequence such that everything beyond fits inside an interval of length ϵ . This immediately clarifies why our previous attempt failed to establish a useful criterion: for a sequence like $c_n = \log(n)$, though it is the case that $|\log(n+1) - \log(n)| \rightarrow 0$ as $n \rightarrow \infty$, it is not the case that $|\log(m) - \log(n)| \rightarrow 0$ for a fixed n if we’re allowed to continue varying m . Concretely, if we take $\epsilon = 1$, then for any proposed thresholding index $N \in \mathbb{N}$, choose $n \geq N$, and then choose $m \geq n$ so large that $m \geq en$ (again, this is possible because $\log(n) \rightarrow \infty$ as $n \rightarrow \infty$). In this case, we have

$$|\log(m) - \log(n)| = \left| \log\left(\frac{m}{n}\right) \right| \geq \left| \log\left(\frac{en}{n}\right) \right| = \log(e) = 1 \equiv \epsilon.$$

Thus, no matter how deep in the tail we go, we can always find two terms whose distance is at least 1.

At this point, we've established the useful criterion that if a sequence a_n converges to a limit a , then the sequence is Cauchy: for any $\epsilon > 0$, there exists a threshold $N \in \mathbb{N}$ such that for any $m, n \geq N$,

$$|a_n - a_m| < \epsilon.$$

Really though, we've yet to establish the desired converse: what we'd like is to be able to say that if a sequence a_n is Cauchy, then it converges to a limit a . The utility in the converse is that whether a sequence is Cauchy can be ascertained completely from the *construction of the sequence itself* and, importantly, ambivalent to a proposal limit a . If we don't know whether a sequence converges, then the matter of proposing a limit to check its convergence is only productive if we have some intuition for what that limit may be. For complicated sequences, this is a difficult task.

What should be somewhat clear, however, is that if a sequence is Cauchy, then it really is the case that the tail of the sequence “gets small and stays small forever.” This is not the complication in establishing that a Cauchy sequence converges to a limit; the complication is with *the limit point itself*. In particular, let us not consider a sequence taking values in all of $\mathbb{R}$, but one which we explicitly dictate to take values in a subset of $\mathbb{R}$: if we let

$$a_n := \frac{1}{n},$$

then clearly $a_1 = 1$, $a_2 = 1/2$, $a_3 = 1/3$, ..., and $a_n \rightarrow 0$ as $n \rightarrow \infty$. However, notice that *none* of the terms $a_n = 0$ exactly; it is only *in the limit* that a_n tends toward $a := 0$. But this means we could very well define the *underlying space* on which the sequence $a_n = 1/n$ is constructed to not be all of $\mathbb{R}$, but just $(0, 1] \subset \mathbb{R}$. Each $a_n = 1/n \in (0, 1]$ for $n \in \mathbb{N}$, so the sequence a_n defined on the space $(0, 1]$ is a completely valid construction. Yet, if we imagine that “the only numbers a we're allowed to name as prospective limits of our sequence a_n are those in $(0, 1]$,” it is no longer the case that a_n converges to a limit in a precise sense; since we've explicitly defined the sequence $a_n = 1/n$ to live on the space $(0, 1]$, and since $a_n \rightarrow 0 \notin (0, 1]$, what we should perceive as happening is our sequence a_n creeps closer and closer to the edge of the world $(0, 1]$, but we're simply “not able to see where it's heading” — even though we can clearly see each member $a_n = 1/n$ in our space $(0, 1]$, its inevitable destination $a = 0$ remains elusive to us.

This example shows something important: whether or not a Cauchy sequence converges to a limit point is not really a fact about sequences alone — rather, it is a fact about the *space the sequence lives in*. The same list of numbers $1, 1/2, 1/3, \dots$ can behave in two *qualitatively* different ways depending on what we “count as a legitimate destination,” even if we're able to *quantitatively* ascertain that the destination “should be the same either way.” In $\mathbb{R}$, the tail of the sequence $a_n = 1/n$ collapses toward 0, and 0 is a real number, so a_n legitimately converges *in* $\mathbb{R}$. On the other hand, when we restrict our space to just $(0, 1]$, the tail of the sequence $a_n = 1/n$ still collapses, but its natural destination lies just beyond the boundary of description relative to the space we've decided to call our home. That is, $a_n = 1/n$ is always a Cauchy sequence, but a_n does *not* converge to a limit in $(0, 1]$, while it *does* converge to a limit in $\mathbb{R}$.

To summarize, a Cauchy sequence a_n is best thought of as a sequence whose tail eventually agrees on where its going. Past some thresholding index $N \in \mathbb{N}$, all of its terms are packed into an arbitrarily small interval; the sequence has essentially committed to a destination. The important question in deciding whether or not to call such a Cauchy sequence convergent is thus a matter of whether that destination is visible to us in the space we've decided to define the sequence over. Intuitively, it should be clear that sequences defined on a world like $(0, 1]$ tend to be a bit messier

than those on $[0, 1]$, simply because we can define sequences like $a_n = 1/n$ whose members all live in $(0, 1]$, but which converge to a point outside $(0, 1]$. This defect is obviously not present when we define $a_n = 1/n$ on $[0, 1]$, since in that case, $a_n \rightarrow 0 \in [0, 1]$.

All of this may seem somewhat pedantic, since our quibble with our ostensibly artificial example $a_n = 1/n$ on $(0, 1]$ is essentially a matter of “just one point, $a = 0$.” For the most part, this is a valid observation, and it is at the heart of the formal workaround which will allow us to say when Cauchy sequences converge to a limit. However, we quickly want to emphasize the fact that just because this example appears artificial does not mean it is a waste of time to ponder its ramifications. A more interesting question might be whether we could have a sequence of real numbers $a_n \in \mathbb{R}$ which converges to a complex-valued limit, say $a = x + iy \in \mathbb{C}$ with nonzero imaginary part $\Im(a) := y \neq 0$. The reason this may not be immediately obvious is because it certainly is the case that sequences of *rational numbers* $a_n \in \mathbb{Q}$, for instance, can converge to *irrational limits* $a \in \mathbb{R} \setminus \mathbb{Q}$. Consider the sequence

$$1, 1.4, 1.41, 1.414, 1.4142, \dots;$$

each member of this sequence is a rational number (we can represent each by a fraction, say $a_1 = 1/1$, $a_2 = 14/10$, $a_3 = 141/100$, and so on). One could construct such a sequence by However, these are clearly just the first few terms in the decimal expansion of $\sqrt{2}$, and we can imagine continuing the construction in such a way that $a_n \rightarrow \sqrt{2}$ as $n \rightarrow \infty$. In this case, it is clear that the sequence a_n is Cauchy (why?), and each member $a_n \in \mathbb{Q}$, but by construction $a_n \rightarrow \sqrt{2} \notin \mathbb{Q}$.

Motivated by this, you may be tempted to try and construct a sequence of real numbers $a_n \in \mathbb{R}$ whose limit $a \notin \mathbb{R}$; as gestured, one natural starting place would be to check whether real sequences could have a complex limit. However, we quickly see this is impossible; since $\mathbb{R}$ forms a line inside the complex plane, and any non-real complex number lies a nonnegligible positive distance above or below that line. Concretely, if $z = x + iy$ with $y \neq 0$, then for every real number r (equivalently, complex number with zero imaginary part), then if we measure the distance from the real r to the complex z by

$$|r - z| = \sqrt{(r - x)^2 + (y)^2} \geq \sqrt{y^2} = |y|.$$

In particular, if we take $\epsilon = |y|/2$, then $|r - z| \geq \epsilon$ for any $r \in \mathbb{R}$. In words, what this means is that the disk of radius $|y|/2$ about the complex point $z = x + iy$ will never contain points on the real-axis. Thus it cannot be that a real sequence a_n converges to a complex limit a .

What we’ve seen thus far are three examples highlighting different convergence phenomena:

- $\mathbb{Q}$ inside $\mathbb{R}$: we can construct a sequence of rational numbers $a_n \in \mathbb{Q}$ which converge to an irrational limit $\sqrt{2} \notin \mathbb{Q}$.
- $(0, 1]$ inside $[0, 1]$: we can construct a sequence of real numbers $a_n \in (0, 1]$ which converge to an exterior point $0 \notin (0, 1]$.
- $\mathbb{R}$ inside $\mathbb{C}$: any convergent sequence of real numbers $a_n \in \mathbb{R}$ will converge to a real limit, so there is *no* “real-to-nonreal” convergence story to tell.

The three examples above display three different flavors of convergence behavior, and we hasten to discuss each in a bit more detail. Again, it is quite simple to find a sequence of rational numbers converging to an irrational limit; we can construct rational approximations to $\sqrt{2}$ using, say, a bisection or Newton’s method applied to $x^2 - 2 = 0$. One can do the same thing for $x^2 - 3 = 0$,

$x^2 - 5 = 0$, and so on. The point we seek to make here is that it is intuitively apparent that “irrational numbers exist,” even if we’re unable to pin them down completely — using a straight-edge to draw the right triangle $a^2 + b^2 = c^2$ with $a = b = 1$ is going to “show you what $\sqrt{2}$ looks like on the hypotenuse $c = \sqrt{2}$,” even though we’re unable to write the decimal expansion of $\sqrt{2}$. Thus, if we “agree to live in the rational numbers $\mathbb{Q}$,” we’ll be forced to live with a “relatively hole-y landscape” — this is somewhat different from the second example, where even though $0 \notin (0, 1]$, the notion of “walking across the landscape $(0, 1]$ ” seems more plausible than “walking across the rational numbers.” The last example, then, shows how a relatively *complete* landscape should feel — there is simply no way we can “chart a set of steps across the real-line $\mathbb{R}$ ” inadvertently leading us *outside* of the real-line $\mathbb{R}$ into, say, the complex numbers $\mathbb{C}$. The subtlety here is that $\mathbb{R}$ can be represented either as its own distinct entity, *or* as a single axis inside the complex plane $\mathbb{C}$, but sequences of real-valued steps a_n will “feel the same” regardless of which world ($\mathbb{R}$ or the real-axis in $\mathbb{C}$) we perceive as truly inhabiting the sequence a_n . This is not the case for rational sequences, where we can “chart sequences of steps leading us outside $\mathbb{Q}$ into $\mathbb{R}$,” or for sequences defined on *open subsets* of the real line, where we can “chart sequences of steps leading us to the edge of the open world.”

It should hopefully be somewhat apparent that $\mathbb{R}$ occupies something of a dignified air amongst the various number systems: sequences we define over *all of* $\mathbb{R}$ seem to stay in $\mathbb{R}$. But we now take one step further back and ask: what *is* $\mathbb{R}$? It’s clear how we can define the rational numbers $\mathbb{Q}$: we can simply take $\mathbb{Q} = \{p/q: p, q \in \mathbb{Z}\}$. But $\mathbb{R}$ is unwieldy by comparison: it includes numbers like $\sqrt{2}$ (which we can define as the positive root of the equation $x^2 - 2 = 0$) but it also includes numbers like π and e , which are somehow “more complicated to describe” than $\sqrt{2}$. Indeed, we cannot define π to be the solution to an algebraic equation as we can $\sqrt{2}$; the same is true for e^2 . We bring this up merely to notice that the real numbers $\mathbb{R}$ cannot be characterized as just “the set of numbers solving some algebraic equation with rational coefficients” — it is more complicated still, containing a class of numbers that clearly characterize tangible objects of interest in practical applications but which remain elusive to our most familiar algebraic methods.

It is perhaps helpful to consider the number e in a bit more detail. There are several ways to define the number e , but before we remind you of them, consider *why* defining such a number would be of import in the first place. That is, if you were “starting from the beginning” and had no knowledge that exponential growth *should* be characterized by the number e , how could you identify and construct it yourself? Again, we remind you of the distinction between *algebraic* numbers — those which we can identify as the roots of algebraic equations with rational coefficients like $x^2 - 2 = 0$ — and *transcendental* numbers, which at this point are (somewhat vaguely) the “non-algebraic real numbers.” But it is somewhat hard to conceive of *why* transcendental numbers should even exist in the first place.

One way to dissolve some of this mystery is to notice that there are really two different notions of “defining a number” floating around in our heads. The algebraic notion is: “give me a finite equation with rational coefficients, and I will point to the real number that solves it.” We can identify $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, and many other irrational reals in this manner. However, it is *analysis* which provides us with a second, equally legitimate notion of constructing real numbers: “give me a *process* that produces better and better *rational approximations*, and I will name the destination of that process.” In this second sense, a number is not just presented to us as a static object we seek to pin down, but as the *limit of a procedure*.

²We call numbers like π and e which are real- (or complex-) valued but which are *not* the solution to an algebraic equation with rational coefficients *transcendental numbers*.

After making this identification, it should be less surprising that $\mathbb{R}$ cannot be pinned down with algebra alone. Indeed, finding roots of a polynomial with rational coefficients is just one kind of rule we can write down, but we know from an introductory exposure that it is the *limit* which is the king of calculus, a fact which can be obfuscated by the time we commit to studying techniques for evaluating derivatives, integrals, and infinite series. But each of these is defined precisely in terms of a limiting procedure: the derivative of a (continuous univariate) function $f(x)$ at a point $x = x_0$ is

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h};$$

the Riemann integral is

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x;$$

and an infinite series is

$$\sum_{i=1}^{\infty} a_i = \lim_{n \rightarrow \infty} \sum_{i=1}^n a_i.$$

A first course in calculus teaches us how to evaluate these limits in some situations, but we often fail to internalize the fact that the limits themselves are often numbers which otherwise defy description! Perhaps the simplest example to demonstrate this is indeed the number e . If we imagine tracking growth of, say, a bank account, bacterial population, or chemical reaction, then we can often imagine the rate of growth of that population as proportional to the size of the population at the time (a bigger bank account accumulates more interest, a larger bacteria colony produces more offspring, etc.). Concretely, if we have a bank account with $x_0 = 1$ unit at baseline accumulating 100% interest every year, then at the end of the first year we will have accumulated

$$x_1 = x_0 \left(1 + \frac{\% \text{ interest}}{100} \right) = 1 \left(1 + \frac{100}{100} \right) = 1(1 + 1) = 2.$$

However, we could also imagine breaking up the interest deposits into six month increments; say, with 50% interest accumulating after the first six months, and another 50% at the end of the year. Then we would have

$$\begin{aligned} x_1 &= x_0 \left(1 + \frac{50}{100} \right) = 1 \left(1 + \frac{1}{2} \right) = 1.5, \\ x_2 &= x_1 \left(1 + \frac{50}{100} \right) = 1.5 \left(1 + \frac{1}{2} \right) = 2.25. \end{aligned}$$

Notice that the computation above is done in a recursive manner; in particular, with $x_0 = 1$, we can compactly find x_2 via

$$x_2 = \left(1 + \frac{1}{2} \right) \left(1 + \frac{1}{2} \right) = \left(1 + \frac{1}{2} \right)^2 = 2.25.$$

Similarly, if we start with $x_0 = 1$ and accumulate 25% interest every quarter (three months), then at the end of the year we would have accumulated a total of

$$x_4 = \left(1 + \frac{1}{4} \right)^4 = 2.441406.$$

In general, we imagine that if we start with $x_0 = 1$ and accumulate $100(1/n)\%$ interest every period, then at the conclusion of n periods, we will have accumulated total value

$$x_n = \left(1 + \frac{1}{n}\right)^n. \quad (1.A7)$$

Something we quickly notice if we try plugging in larger and larger values of n into (1.A7) is that it never seems to exceed 3; for example:

```
xn <- function(n) (1 + 1/n)^n  
xn(1) # Annually
```

```
[1] 2
```

```
xn(2) # Bianually
```

```
[1] 2.25
```

```
xn(4) # Quarterly
```

```
[1] 2.441406
```

```
xn(12)
```

```
[1] 2.613035
```

```
xn(120)
```

```
[1] 2.707041
```

```
xn(1200)
```

```
[1] 2.71715
```

```
xn(12000)
```

```
[1] 2.718169
```

```
xn(120000)
```

[1] 2.718271

```
xn(1200000)
```

[1] 2.718281

Thus, what we observe is that, for instance,

$$\left(1 + \frac{1}{12}\right)^{12} \approx 2.613, \quad \left(1 + \frac{1}{120}\right)^{120} \approx 2.707, \quad \left(1 + \frac{1}{1200}\right)^{1200} \approx 2.717,$$

and by the time n grows quite large, we can observe that the sequence seems to settle down; in particular, it appears as if

$$\left(1 + \frac{1}{n}\right)^n \xrightarrow{n \rightarrow \infty} 2.71828\dots$$

Again, what's important about this is that the number 2.71828... seems to have materialized not from an algebraic equation, but as a result of our defining a *limiting process* and observing where it seems to be headed. The question now is whether we can formally characterize this limiting behavior; i.e., whether it *really is* the case that $x_n := (1 + 1/n)^n < \infty$ for all $n \in \mathbb{N}$, and if it is, whether we can identify a concrete limiting value (i.e., we must not discount the possibility that x_n behaves in an erratic, unpredictable manner for different magnitudes of $n \in \mathbb{N}$ — it may be the case that x_n *appears* to settle down towards a limit, but that after some threshold a new, possibly unexpected behavior begins to emerge).

Going back to when we plugged in larger and larger values to the sequence $x_n = (1 + 1/n)^n$, we can observe two critical heuristics in establishing the existence of a limit $\lim x_n$:

- (i) **Monotonicity:** It appears as if x_{n+1} is larger than x_n no matter the $n \in \mathbb{N}$ we decide to plug in.
- (ii) **Boundedness:** No matter how large an n we plug in to x_n , it appears as if x_n never exceeds, say, the number 3.

Thinking purely from our intuition, a monotone sequence is one which avoids the unpredictability we alluded to earlier; that is, we can expect consistent growth or decay of a monotone sequence as we take larger and larger n , without the possibility of “unexpected turn-arounds.” It is not immediately obvious that the sequence under investigation $x_n = (1 + 1/n)^n$ is monotone; to see this, recall the *binomial formula*, which says that we may expand a binomial $(x + y)^n$ as

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}.$$

(Notice that we can always flip the positions of a and b in the sum on the left, so that either a or b can be written as powers of k or $n - k$ depending on which is more convenient.) Now, noticing that our sequence is of this form, we can write

$$\left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^n \binom{n}{k} \left(\frac{1}{n}\right)^k = \binom{n}{0} \left(\frac{1}{n}\right)^0 + \binom{n}{1} \left(\frac{1}{n}\right)^1 + \binom{n}{2} \left(\frac{1}{n}\right)^2 + \cdots + \binom{n}{k} \left(\frac{1}{n}\right)^k + \cdots$$

We can simplify the binomial coefficients as

$$\binom{n}{0} \left(\frac{1}{n}\right)^0 + \binom{n}{1} \left(\frac{1}{n}\right)^1 + \binom{n}{2} \left(\frac{1}{n}\right)^2 + \cdots + \binom{n}{k} \left(\frac{1}{n}\right)^k + \cdots = 1 + n \left(\frac{1}{n}\right) + \frac{n(n-1)}{2!} \frac{1}{n^2} + \cdots + \frac{n(n-1) \cdots (n-k+1)}{k!} \frac{1}{n^k} + \cdots$$

and, if we look at only the lattermost general term, we can write

$$\frac{n(n-1) \cdots (n-k+1)}{k!} \frac{1}{n^k} = \frac{1}{k!} \underbrace{\left(\frac{n}{n}\right)}_{=1} \left(\frac{n-1}{n}\right) \cdots \left(\frac{n-k+1}{n}\right) = \frac{1}{k!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right).$$

Again, this is the k th term in the binomial expansion, so

$$\left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^n \frac{1}{k!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right). \quad (1.A8)$$

Now, if we imagine replacing n by $n + 1$, what we have is

$$\left(1 + \frac{1}{n+1}\right)^{n+1} = \sum_{k=0}^{n+1} \frac{1}{k!} \left(1 - \frac{1}{n+1}\right) \left(1 - \frac{2}{n+1}\right) \cdots \left(1 - \frac{k-1}{n+1}\right) \quad (1.A9)$$

Notice that each (1.A8) and (1.A9) are the product of numbers of the form $(1 - \text{small positive number}) < 1$, but that the terms in (1.A8) are each slightly smaller than their corresponding terms in (1.A9) since (for example)

$$\frac{1}{n} > \frac{1}{n+1} \implies 1 - \frac{1}{n} < 1 - \frac{1}{n+1}.$$

Each of the $k = 0, 1, \dots, n$ terms in (1.A9) will thus be slightly larger than the corresponding terms in (1.A8). Furthermore, there is an *additional* positive $n + 1$ term in (1.A9) not present in (1.A8); it is thus clear that (1.A9) sums more terms, each of which is slightly larger than the corresponding terms in (1.A8) — we thus conclude that, indeed,

$$\forall n \in \mathbb{N}: x_{n+1} = \left(1 + \frac{1}{n+1}\right)^{n+1} > \left(1 + \frac{1}{n}\right)^n = x_n.$$

We've just shown that the sequence $x_n = (1 + 1/n)^n$ is *monotone increasing*; its terms $x_{n+1} > x_n$ for all $n \in \mathbb{N}$. There are really two possibilities: either $x_n \rightarrow \infty$ as $n \rightarrow \infty$, or $x_n \rightarrow x < \infty$ for some number x . From our earlier inspection, it already seems likely that x_n converges, since on plugging in larger and larger n we were unable to find an x_n exceeding the number 3. However, to

be rigorous, we can pursue an argument in a manner similar to the above: we already observed in (1.A8) that

$$\left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^n \frac{1}{k!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right),$$

where each term in the sum is comprised of products of numbers strictly less than 1. That is,

$$\frac{1}{k!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right) < \frac{1}{k!},$$

since the left hand side is $1/k!$ multiplied by a number less than 1. Since this inequality holds for each term in the sum, it will also hold over the sum itself; in particular,

$$x_n = \left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^n \frac{1}{k!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right) < \sum_{k=0}^n \frac{1}{k!}.$$

Now, from

$$x_n < \sum_{k=0}^n \frac{1}{k!} = 1 + 1 + \frac{1}{2!} + \cdots + \frac{1}{n!},$$

observe that for each $k \geq 2$,

$$k! = k(k-1) \cdots (2)(1) \geq 2(2) \cdots (2)(1) = 2^{k-1},$$

where we're just "replacing the larger integers $k, k-1, \dots$ on the left with 2s on the right." This means that

$$x_n < \sum_{k=0}^n \frac{1}{k!} \leq 1 + \left(\frac{1}{2^0} + \frac{1}{2^1} + \frac{1}{2^2} + \cdots + \frac{1}{2^{n-1}}\right) = 1 + \sum_{k=0}^{n-1} \frac{1}{2^k}.$$

To summarize, the reason we've argued in this manner is precisely because we don't have a *closed-form expression* for evaluating the sums we've used to bound our original binomial expansion,

$$\left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^n \binom{n}{k} \left(\frac{1}{n}\right)^k.$$

Binomial coefficients are tough to work with in this form, so we expanded the sum as

$$\left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^n \frac{1}{k!} \left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \cdots \left(1 - \frac{k-1}{n}\right).$$

Still, we don't have a clean way to evaluate the sum on the right without computing all n terms in the sum individually and combining them at the end — if our goal is to show that the sum is bounded for *all* $n \in \mathbb{N}$, the form above will be insufficient. However, we don't need to evaluate the sum itself to show it is bounded; if we can compare it to a sum which is strictly larger for any $n \in \mathbb{N}$ and show that *that* sum is bounded, the boundedness of the original will follow by monotonicity. (Notice that monotonicity is necessary for this argument to work as described; otherwise, it could be that the original sum is strictly less than our comparator for all $n \in \mathbb{N}$, but that the original reverses course and begins to decrease after some n .)

In pursuit of this, we established that

$$\left(1 + \frac{1}{n}\right)^n < \sum_{k=0}^n \frac{1}{k!},$$

which certainly appears to be a simpler sum to work with. The last maneuver is perhaps the most surprising one; rather than working directly with the sum above, we instead bounded it by

$$\sum_{k=0}^n \frac{1}{k!} \equiv 1 + \sum_{k=1}^n \frac{1}{k!} < 1 + \sum_{k=0}^{n-1} \frac{1}{2^k}.$$

Now, it may not be entirely evident why bounding $\sum_k 1/k!$ by $\sum_k 1/2^k$ is in any way helpful, but let us realize *why* the former is “difficult,” and why the latter is somewhat “easy” by comparison. Define

$$S_n := \sum_{k=0}^n \frac{1}{2^k} \equiv 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots + \frac{1}{2^n}.$$

If we peel off the first $k = 0$ term from S_n , what we’re left with is

$$\frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^n} = \frac{1}{2} \left(1 + \frac{1}{2} + \frac{1}{4} + \cdots + \frac{1}{2^{n-1}} \right) = \frac{1}{2} S_{n-1},$$

so that, in particular, we can establish the recursive relationship between S_n and S_{n-1} for any $n \in \mathbb{N}$:

$$S_n = \frac{1}{2} S_{n-1} + 1. \tag{1.A10}$$

Realize, however, that *in the limit* as $n \rightarrow \infty$, both S_n and S_{n-1} describe the same infinite series — namely,

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} S_{n-1} = \sum_{k=0}^{\infty} \frac{1}{2^k}. \tag{1.A11}$$

This is quite an important realization. Defining

$$S := \lim_{n \rightarrow \infty} S_n = \sum_{k=0}^{\infty} \frac{1}{2^k} \tag{1.A12}$$

and recalling that for a convergent sequence $x_n \rightarrow x$ as $n \rightarrow \infty$ and real constants $a, b \in \mathbb{R}$, we have

$$\lim_{n \rightarrow \infty} ax_n + b = a \left(\lim_{n \rightarrow \infty} x_n \right) + b = ax + b,$$

what (1.A10) and (1.A11) imply is that

$$\lim_{n \rightarrow \infty} S_n = \frac{1}{2} \left(\lim_{n \rightarrow \infty} S_{n-1} \right) + 1,$$

or

$$S = \frac{1}{2} S + 1 \implies S = \frac{1}{1 - \frac{1}{2}} = 2.$$

In particular, notice that the first term in the sum (1.A12) is $a = 1$, and that each successive term differs from the previous by a *common ratio* of $r = 1/2$. We thus might think it sensible to, in general, claim that

$$S = a + rS \implies S = \frac{a}{1-r}.$$

The natural question is for which values of $r \in \mathbb{R}$ and $a \in \mathbb{R}$ this holds true — we claim it holds for $a = 1$ and $r = 1/2$, but is it always the case?

We take this opportunity to present a general definition.

Definition 1.0.3 (Geometric Series)

Let

$$S_n := a + ar + ar^2 + \cdots + ar^n = \sum_{k=0}^n ar^k.$$

Then, for any ratio $r \in \mathbb{R}$ and initial point $a \in \mathbb{R}$, the value of S_n for a given $n \in \mathbb{N}$ is

$$S_n := \sum_{k=0}^n ar^k = \frac{a(1-r^{n+1})}{1-r}.$$

Furthermore, suppose that $-1 < r < 1$ strictly. Then

$$S = \lim_{n \rightarrow \infty} S_n \text{ exists,}$$

and

$$S = \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \sum_{k=0}^n ar^k := \sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}; \quad -1 < r < 1, \quad a \in \mathbb{R}. \quad (\text{Geometric Series})$$

Remark.

Notice that the *partial sum* of the geometric series is always well-defined (i.e., for any ratio $r \in \mathbb{R}$) — its value is

$$S_n = \sum_{k=0}^n ar^k = \frac{a(1-r^{n+1})}{1-r}, \quad \forall r \in \mathbb{R}.$$

On the other hand, we only define the *limit* for $|r| < 1$ — i.e.,

$$\lim_{n \rightarrow \infty} S_n = \sum_{k=0}^{\infty} ar^k = \frac{a}{1-r} \iff -1 < r < 1.$$

The reason for this is readily apparent when we take the limit relative to the closed-form expression for the partial sum. What we have is

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \frac{a(1 - r^{n+1})}{1 - r} = \frac{a(1 - \lim_{n \rightarrow \infty} r^{n+1})}{1 - r}.$$

One then observes that

$$\lim_{n \rightarrow \infty} r^{n+1} = \begin{cases} 0, & -1 < r < 1, \\ \infty, & r \geq 1, \\ \text{DNE}, & r \leq -1. \end{cases}$$

We say that the limit is ∞ in the case where $r > 1$, but that the limit *does not exist* whenever $r < -1$. The reason for this is that when $r > 1$, $r^n < r^{n+1}$ for any $n \in \mathbb{N}$; on the other hand, when $r < -1$, the powers r^n differ in sign depending on whether n is even or odd! It is thus the case that r^n is monotonic but unbounded for $r > 1$, but is neither monotonic nor bounded for $r < -1$. Since it would seem we "know a bit more about where the sequence is headed" in the former case, it is common to distinguish these cases by using the symbols $\pm\infty$ whenever a sequence is monotonic but unbounded, and to use DNE whenever some sort of *periodic* behavior is displayed by a sequence. (We'll formalize this notion using the lim sup and lim inf in the coming sections.)

The example we've just gone through illustrates the evaluation of a geometric series with $a = 1$ and $r = 1/2$ — that is,

$$S = \sum_{k=0}^{\infty} \frac{1}{2^k} = \frac{a}{1 - r} = \frac{1}{1 - \frac{1}{2}} = 2.$$

What this means in the context of our current derivation is that

$$\left(1 + \frac{1}{n}\right)^n < 1 + \sum_{k=0}^{n-1} \frac{1}{2^k},$$

so that, in the limit as $n \rightarrow \infty$,

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n < 1 + \lim_{n \rightarrow \infty} \sum_{k=0}^{n-1} \frac{1}{2^k} = 1 + \lim_{n \rightarrow \infty} S_{n-1} = 1 + S = 1 + 2 = 3.$$

What we've just shown is that, for the sequence

$$x_n = \left(1 + \frac{1}{n}\right)^n,$$

we have both

$$x_n < x_{n+1}, \quad \forall n \in \mathbb{N}, \quad (x_n \text{ is Monotone Increasing})$$

and

$$\lim_{n \rightarrow \infty} x_n < 3. \quad (x_n \text{ is Bounded})$$

Again, what we're trying to do here is to evaluate $\lim_{n \rightarrow \infty} (1 + 1/n)^n$. We used the binomial formula to write this as an infinite series, but since evaluating infinite series is quite difficult, we wanted to check whether it even makes sense that the series would converge to a finite value as $n \rightarrow \infty$. Having seemed to confirm this (or at least having not shown the contrary to be obvious), what we'd like to do ideally is find the actual value of this limit — i.e., to find the value of the limit. Let us agree to call this limit e — i.e., we *define* the number e to be

$$e := \lim_{n \rightarrow \infty} x_n;$$

that is, we can define e to be any of

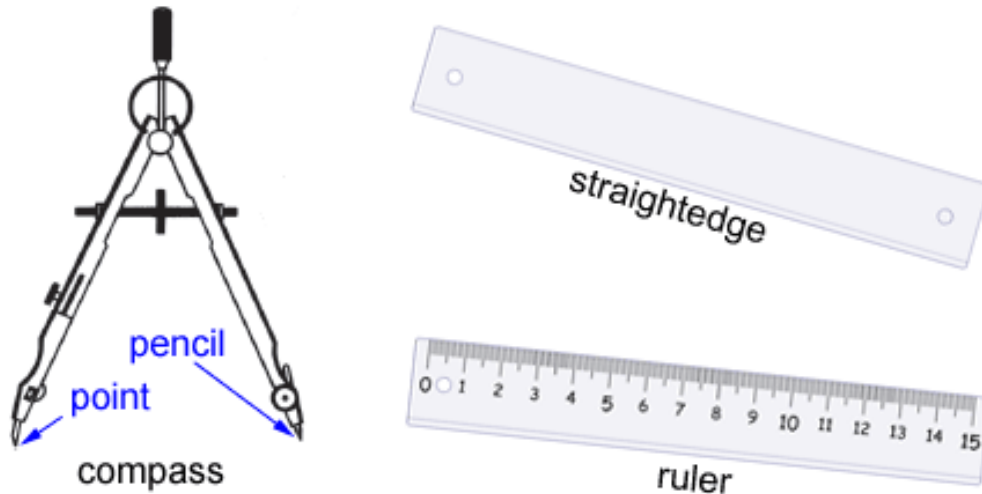
$$e := \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{1}{k!} \equiv \sum_{k=0}^{\infty} \frac{1}{k!}. \quad (\text{Definition of } e)$$

At this point, we urge you to cast aside any pre-existing notions of “what the number e is supposed to be” — you almost certainly know it to be $e \approx 2.71828\dots$, but this is not the point. It is our observation that, often, it feels as though numbers like e are “handed from on high”; as though the decimal 2.71828 is the “object of significance.” But again, we defined e to be the *limit of a process* — that this limit “happens to admit a decimal expansion” is better understood as a useful coincidence rather than the fundamental object that e intends to describe. The reason we hasten to make the distinction is because a number's decimal representation seemingly falls into one of two camps — it is relatively common knowledge that a rational number's decimal expansion terminates in some sort of identifiable pattern, while an irrational number's does not.

Yet, while this binary rational/irrational split is a useful heuristic for describing real numbers, it is occasionally a bit of an oversimplification. For one, we've already observed that both $\sqrt{2}$ and e are irrational numbers, even though $\sqrt{2}$ arises explicitly as the unique positive root of the algebraic equation $x^2 = 2$, while $e = \sum_{k=0}^{\infty} 1/k!$ arises more-or-less from our own intuition on the limiting behavior of convergent sequences. Perhaps more concretely, suppose you were asked to draw the number $\sqrt{2}$ on the usual 2D Euclidean plane; how would you do it? Well, because $\sqrt{2}$ is the unique positive root to the algebraic equation $x^2 = 2$, we can use the Pythagorean formula: the right triangle whose opposite and adjacent sides are of unit length $a = b = 1$ has hypotenuse of length c satisfying

$$a^2 + b^2 = c^2 \implies c^2 = 1^2 + 1^2 = 2 \implies c = \sqrt{2}.$$

In this way, we can imagine that $\sqrt{2}$ is somewhat tangible in spite of its irrationality — given only a *straightedge ruler* and a *compass* (see the figure below), we're able to construct the length $\sqrt{2}$ in exactly three physical moves (draw $a = 1$, a perpendicular $b = 1$, and connect the ends). In general, we can pursue an algorithm like the following to construct a length $\sqrt{x}$ for some given length $x > 0$:



Straightedge-and-Compass (<https://www.math.net/geometric-construction>)

- (i) Assume we start with a unit segment of length 1 and the positive length $x > 0$. On a line, mark three points A, B, C such that $AB = 1$ and $BC = x$; hence $AC = 1 + x$.
- (ii) Using the compass, draw the semicircle with diameter AC .
- (iii) Draw the perpendicular through B and let D be the point this perpendicular intersects the semicircle.
- (iv) Then, the new segment BD is of length $\sqrt{x}$.

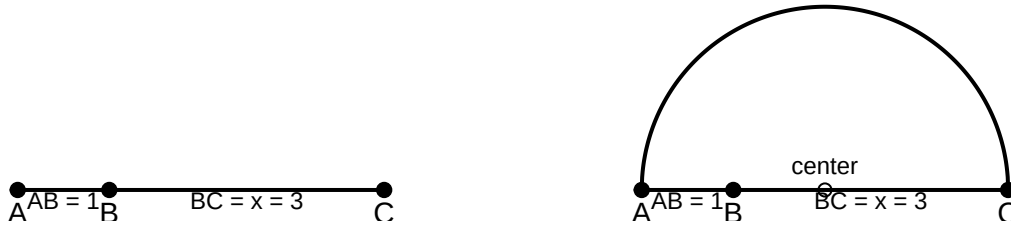
This algorithm arises simply from the self-similarity of the triangles constructed in this manner. In particular, $\triangle ADB$ is a right triangle, and the *altitude* BD splits this big right triangle into two smaller right triangles $\triangle ABD$ and $\triangle CBD$, each similar to the original. A fact about similar triangles is that their corresponding sides must satisfy the proportionality relations

$$BD^2 = AB \cdot BC;$$

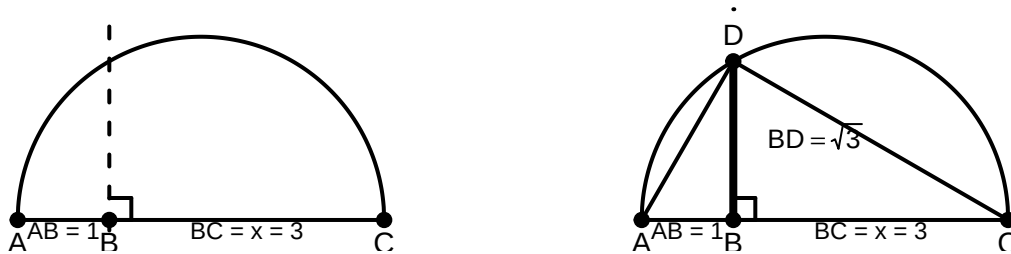
whence, if $AB = 1$ and $BC = x$, then $BD^2 = 1 \cdot x$ and

$$BD = \sqrt{1 \cdot x} = \sqrt{x}.$$

Step 1: Mark $AB = 1$ and $BC = x$ on a line **Step 2: Draw the semicircle with diameter**



Step 3: Erect the perpendicular through **Step 4: Intersection gives $BD = \sqrt{x}$**

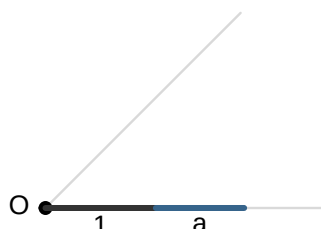


Thus, we've illustrated a concrete, physical manner in which we can construct a segment of length $\sqrt{x}$ for any positive $x > 0$ in the usual 2D Euclidean plane. The point, though, isn't really to convince you that "the number $\sqrt{2}$ exists" — this is already intuitively obvious *because* we can describe $\sqrt{2}$ algebraically as the unique positive root to the equation $x^2 = 2$ over the real line $\mathbb{R}$. The point is more so that we can actually construct a representation for the *physical magnitude* of $\sqrt{2}$ through a *finite process* of construction. In some way, this makes the existence of $\sqrt{2}$ feel grounded in the same Euclidean reality as the number 1.

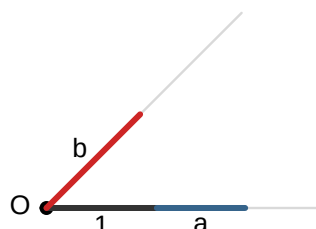
This tangibility extends beyond the extraction of roots. If we accept the existence of a unit length 1, we can effectively turn the plane into something of an arithmetic engine. Concretely, realize that though we hadn't formally defined the concept in a Euclidean sense, the construction above effectively describes an intuitive means for *multiplication*: if $AB = 1$ is of unit length and we use our straightedge and compass to draw a point C a distance of x units to the right of B so that $BC = x$, then by drawing the semicircle connecting the endpoints A and C , the point D where the perpendicular erected through AB at the point B hits that semicircle is of length $\sqrt{1 \cdot x}$, or $\sqrt{AB \cdot BC}$.

To multiply line segment a by segment b , we first construct a reference triangle using our unit length. We lay out the unit 1 and the length a along our horizontal ray, and mark the length b along a second, angled ray. By connecting the unit point 1 to the point b on the second ray, and then constructing the *parallel* to that line passing through a , the new point of intersection on the ray defined by extending b in the same direction will be of length ab :

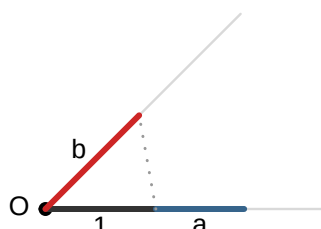
Step 1: Mark Unit 1 and Length a



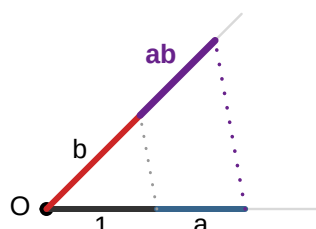
Step 2: Mark Length b on second ray



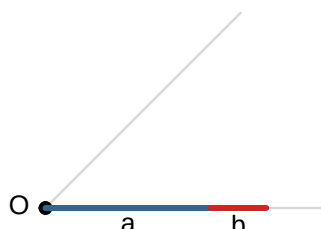
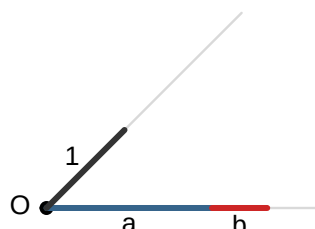
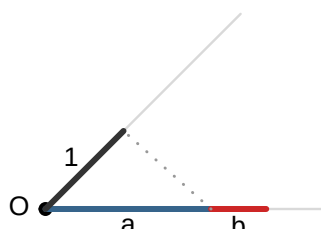
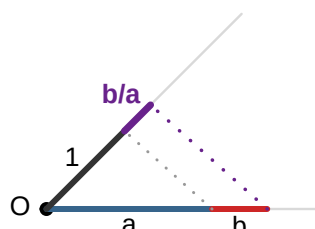
Step 3: Establish Proportionality (1 to b)



Step 4: Parallel through a yields ab



The process for division is the precise inverse of multiplication. If we wish to find b/a , we begin with the divisor a and the numerator b on the primary ray, and mark our unit 1 on the second. By connecting the divisor a back to the unit 1, then constructing the parallel passing through the end of the segment b , the intersection with the ray defined by extending the unit length 1 will be of length b/a :

Step 1: Mark Divisor a and Numerator b**Step 2: Mark Unit 1 on second ray****Step 3: Connect Divisor a to Unit 1****Step 4: Parallel through b yields b/a**

What these constructions reveal is that arithmetic is not a magical operation that happens “to” numbers; it is a *scaling transformation*. By anchoring our triangles at the origin, we see that the segment ab is simply the segment b stretched by the factor a . This realization provides something of a useful reassurance: if we can multiply, divide, and take square roots using nothing but a straightedge and compass, then the entire field of *constructible numbers* — the numbers reachable through finite algebraic steps — should be tangibly accessible to us in our physical reality.

But let us pause for a moment and reflect on the straightedge-and-compass construction of $\sqrt{2}$ above. At some level, it can be easy to lose sight of all the various interconnecting pieces: on the one hand, $\sqrt{2}$ is an irrational number — we will even prove this formally via a classical argument, but for now we presume it to be true. Thus, we can *never* hope to pin down the decimal expansion of $\sqrt{2}$ exactly and, in this way, the number $\sqrt{2}$ is and always will be elusive to us. Yet, viewed in a different way, clearly we can *see* the number $\sqrt{2}$, and we don’t even need the full altitude construction we gave above — all we need is a unit length 1 perpendicular to another, and the hypotenuse of the resulting triangle is *guaranteed* to be of length $\sqrt{2}$. How can a number be too slippery to pin down yet seemingly remain accessible to us physically?

The resolution is that we’ve glossed over a very fundamental property of straightedge-and-compass constructions in our casual description above. Indeed, one can *not* simply presume “a unit length placed perpendicularly to another” — we must *construct* such a perpendicular, and this is not something we can accomplish with the straightedge alone. Indeed, the straightedge is a tool for drawing a line through two *already-known* points, but it has no mechanism for producing a genuinely new point by itself. (Realize that a straightedge is different from a ruler — a ruler is marked with units, which are born from a number-construction process such as this one, thus

requiring the subsequent development to be sensibly defined). That is, if we are given only a single segment AB , then the straightedge can certainly extend the line through A and B , but it cannot conjure any third point C off that line in any principled way. We can plot points arbitrarily all we like, but if we hope for those points to have tangible meaning, what we require is the compass.

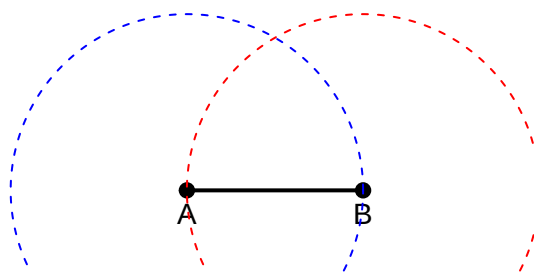
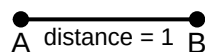
The compass is not merely a “circle drawing device” in some ornamental sense; it is the mechanism by which we impose *distance constraints*. To see what we mean, recall what a circle *is*: given a center point P and a radius $r > 0$, the circle of radius r centered about P is precisely the set of all points Q in the 2D plane satisfying

$$\text{dist}(P, Q) = r. \quad (\text{Circle of radius } r \text{ centered about } P)$$

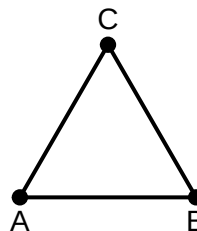
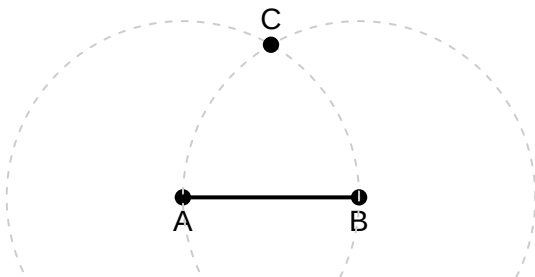
Thus, if we start with points scattered across a 2D Euclidean plane, the straightedge is what allows us to connect those points linearly, while the compass is what allows us to materialize new points we haven’t seen before, assuring us these new points satisfy the principled constraint that “all are the same distance from our chosen center.” When we draw a circle, we are not merely decorating the plane; we are adding a *measuring device* converting the featureless plane into a landscape where distances between arbitrary points become meaningful and reproducible.

To see how this “measuring device” actually functions to expand our world, consider the most basic construction: the equilateral triangle. If we are given only two points A and B , we can use the straightedge to connect the two and form the segment AB , which we might (somewhat arbitrarily) decide to call our “unit length” 1. At this stage, our world is strictly one-dimensional: we only have the line passing through A and B . To move into 2D in a principled way, we set the compass to the distance AB , then draw two circles of radius $r = AB$ about each point A/B . What we then have are circles with centers A and B , each of radius r ; however, more importantly, we’ve implicitly defined two sets of distance constraints in our 2D plane. That is, after drawing the two circles of the same fixed radius 1 about A and B , what we know is that the *points of intersection* between the two circles will be those points in the 2D plane which are simultaneously a unit length away from the center A and a unit length from the center B . There will clearly be two such points of intersection; if we label the upper point C , then what we know is the lengths $AC = BC = 1$, and that $AB = 1$. But this is precisely one definition of an equilateral triangle — i.e., $\triangle ABC$ has equal sides $AB = BC = AC = 1$, so it must be equilateral.

Step 1: Mark A and B, form unit segment AB: Draw circles of radius AB centered at A



Step 3: Identify intersection point C Step 4: Connect AC and BC to form triangle ABC



Notice that nowhere in this construction did we mention the notion of “angle” — that is, we must realize that an equilateral triangle is equivalently defined in terms of either “equal angles” *or* “equal side lengths,” and that satisfying either constraint naturally gives rise to the other. This is not obvious from our construction, though; how can we prove it?

We first must realize that we haven’t actually formally defined a way to measure angles using only a straightedge-and-compass — that is, nowhere in this construction did we mention a protractor, degrees, or any numerical “angle measure.” However, also realize that the fundamental importance here is *not* the “specific value” of an angle in degrees or radians, but rather in the *congruence* of pairs of angle — though we lack a well-defined means to measure angles, we can assert their equivalence (congruence, written $\cong$) from the straightedge-and-compass alone.

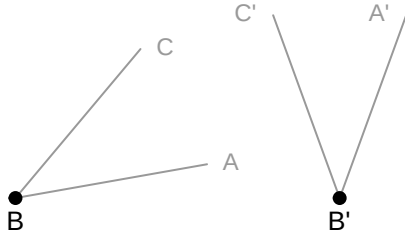
What we claim is that $\angle ABC \cong \angle BCA \cong \angle CAB$. To prove this, start by realizing that it is certainly the case that $\triangle ABC \cong \triangle BCA$: this is the case since, if we consider the corresponding sides of the two triangles (i.e., if we say that the bottom side of the triangle is described by the first two letters so that $\triangle ABC$ has bottom side AB and sim. $\triangle BCA$ with bottom side BC , we imagine placing these two triangles next to one another and comparing their side lengths), we have that $AB = BC = 1$, $BC = CA = 1$, and $CA = AB = 1$. Because all pairs of corresponding sides are of equal length, it would intuitively seem clear these are the “same triangle” — indeed, by *SSS Congruence* (Side-Side-Side congruence), this is sufficient to establish that $\triangle ABC \cong \triangle BCA$. Because “Corresponding Parts of Congruent Triangles are Congruent” (often abbreviated to *CPCTC* in proofs), it follows that $\angle ABC \cong \angle BCA$. Now, the exact same kind of argument can be used to establish that $\triangle ABC \cong \triangle CAB$, leading to $\angle ABC \cong \angle CAB$. Together, what we have is

$$\angle ABC \cong \angle BCA \cong \angle CAB,$$

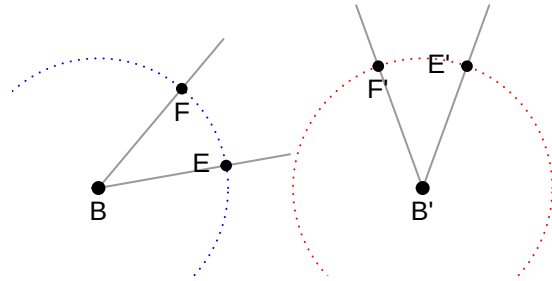
which is exactly what we'd sought to prove.

This brings us to a more general question of how we can compare angles scattered arbitrarily across the 2D plane in general. That is, imagine being presented with two arbitrary angles $\angle ABC$ and $\angle A'B'C'$ scattered across the plane like loose sticks. To determine if the angles are congruent without a protractor to tell us the specific measures of the angles, we can pursue an approach as follows: setting our compass to some fixed radius $r > 0$, draw the circles respectively centered at B and B' . Proceed to mark the points of intersection of the circles with the rays A/A' as E/E' , and the points of intersection with B/B' as F/F' . This creates two isosceles triangles $\triangle BEF$ and $\triangle B'E'F'$, where we have *already guaranteed* that two sides of the first triangle match two sides of the second ($BE = B'E' = r$ and $BF = B'F' = r$). The third sides described by the *chords* EF and $E'F'$ are the only variables left — if it were the case that these sides were congruent, then we would have $BE \cong B'E'$, $BF \cong B'F'$, and $EF \cong E'F'$, meaning we'd be able to use SSS congruence along with CPCTC to establish that $\angle ABC \cong \angle A'B'C'$.

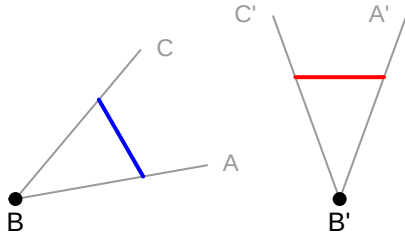
Step 1: Two angles ABC and A'B'C'



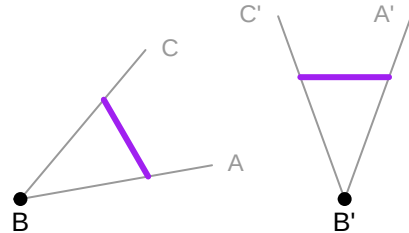
Step 2: Mark points E, F at fixed radius r



Step 3: Connect chords EF and E'F'



Step 4: Congruence established via chord

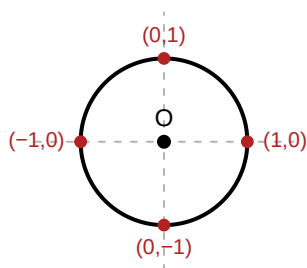


The reason for our detour into straightedge-and-compass constructions is not to belabor their technical aspects, but to highlight a fundamental tension in how we describe the world. To Euclid, the *circle* is a primary, irreducible motion — a singular “swing” of the arm that enforces a distance constraint. Yet, when we attempt to bring algebra back into the fray, we’re asserting a *rectangular grid* as the “default language” for describing points in space. Thus, when we attempt to place our circle onto a Cartesian coordinate plane, we are forced to translate that fluid, continuous *motion* back into the rigid language of horizontal and vertical *distances*.

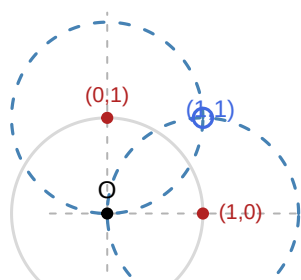
What we might imagine is that, realizing we’ve been arbitrarily defining a “unit length” in our previous straightedge-and-compass constructions, we start by making the principled move of once-

and-for-all setting a unit length $r = 1$ and dictating a point in space O to be the *origin point* $(x, y) = (0, 0)$ of the ambient space (perhaps this point is placed near the middle of a sheet of paper). With our compass set to the unit length $r = 1$, draw a circle about the origin O , use a straightedge to draw a line through this circle (this will be one of our 2D coordinate axes), then construct the perpendicular to this line (this will be the other axis). If we define the points of intersection of the perpendicular axes with the circle of radius $r = 1$ centered at O , what we'll have are four such points a unit length away from O in all four cardinal directions — i.e., we'll have points at $(1, 0)$, $(0, 1)$, $(-1, 0)$, and $(0, -1)$. We can repeat this process, taking these new points as centers of circles of unit radius, constructing perpendiculars, and identifying the intersections of the perpendiculars with the circles to dictate the full plane.

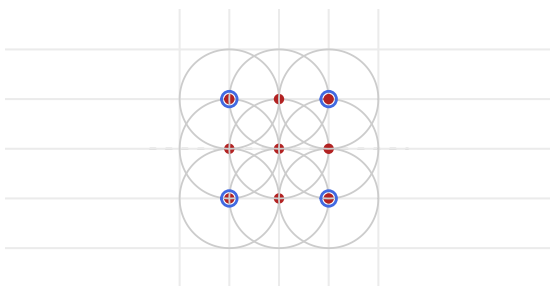
Step 1: Fix the origin and unit radius



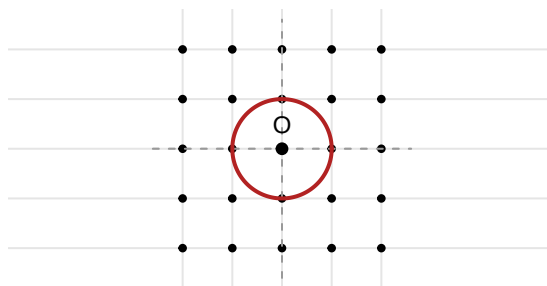
Step 2: Dictate new points



Step 3: Repeating the process



Step 4: Completing the grid

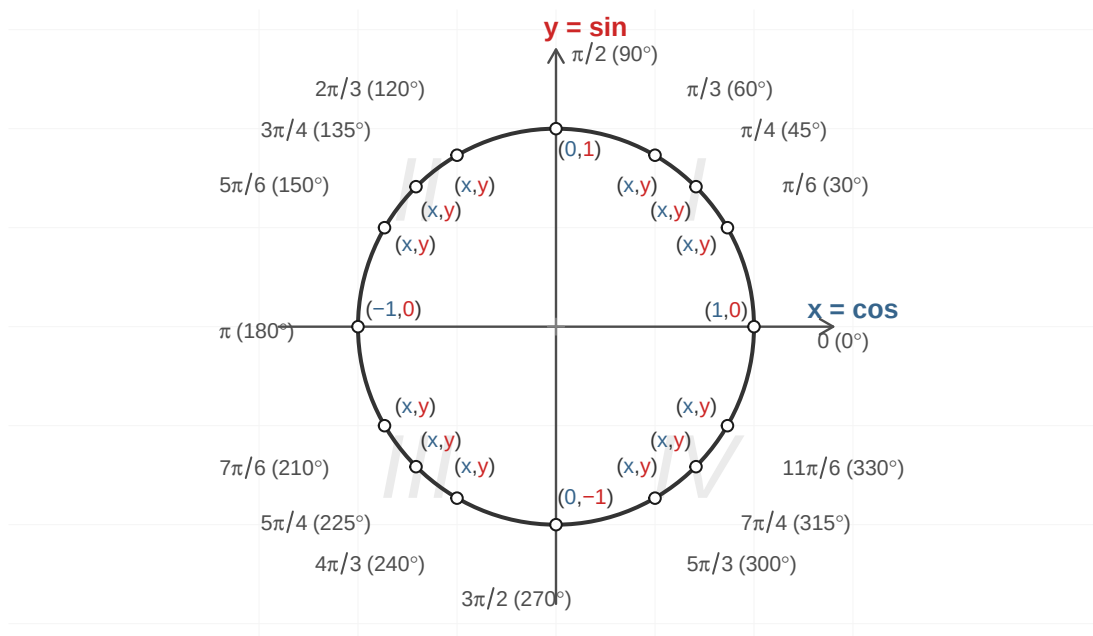


The grid we've constructed above allows us to name points like $(1, 0)$ and $(1, 1)$ relative to our given unit length with absolute precision. But notice that the only points we've truly pinned down in our construction process are pairs of integers $\{(\pm m, \pm n) : m, n \in \mathbb{Z}\}$ — that is, even though it's visually clear that numbers like $(1.5, 1.5)$ are present in the grid, we've not identified them explicitly. Of course, identifying *rational* points like $(1/2, 1/2)$ is a simple matter of further bisection; we omit the details here³, but it is essentially true that we can reach any rational number with these sorts of constructions, and indeed *some* irrationals. One such irrational is clearly $\sqrt{2}$, as we've already shown — if we're given a unit length, then erecting a perpendicular of the same unit length at and endpoint of the original segment will result in a triangle whose hypotenuse is of length $\sqrt{2}$. Combined with the fact that we can multiply and divide numbers, it would then seem sensible

³An excellent source for more on Euclidean constructions is the following: Stillwell, John. *The Four Pillars of Geometry*. Springer, 2005.

that a number like $\sqrt{2}/2$ would be constructible using the straightedge-and-compass. That is, if we know that an irrational number is constructible, then we immediately gain access to countably many new irrationals by scaling and translating by non-zero rationals. The question, then, would seem to be: which irrationals *can* we construct, and how?

The Unit Circle



The Unit Circle (Labelled)

